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Detection of Cryptocurrency Pump-and-Dump Schemes: A Multi-Resolution Benchmark of Five Supervised Classifiers

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Abstract

Cryptocurrency pump-and-dump schemes are coordinated market manipulations in which groups inflate low-volume assets before selling at the peak, causing losses for uninformed traders. This project develops a real-time detection framework using Binance market microstructure signals and extends La Morgia et al. (2021) by benchmarking five supervised classifiers Naive Bayes, Random Forest, XGBoost, Linear SVM, and a focal-loss DNN across 5, 15, and 25-second resolutions on 327 confirmed Telegram-coordinated pump events. We address severe class imbalance, event-level leakage, and misleading ROC-AUC scores by using event-disjoint train/validation/test splits, validation-based threshold tuning, PR-AUC/F1-focused evaluation, and a single audited test read. The selected Random Forest at 15s achieved test F1 = 0.883, precision = 0.930, and recall = 0.841 at threshold 0.66. A 1,000-iteration event-level bootstrap produced a 95% F1 confidence interval of [0.818, 0.943], placing the prior reported F1 of 0.945 within statistical range. GroupKFold validation and a chronological retrain further confirmed stability. Overall, our results show parity with prior work under stricter evaluation rather than a clear improvement, and demonstrate that PR-AUC is more informative than ROC-AUC under 1:1500+ class imbalance. exceeded 0.998 for every model including the worst performer (Naive Bayes, F1 = 0.21 at best).

1. Introduction

1.1 Background

Cryptocurrency markets are less regulated than traditional equities, making them vulnerable to coordinated manipulation. In pump-and-dump schemes, organizers announce a target coin and execution time through Telegram or Discord groups, triggering simultaneous buying that inflates the price before organizers sell at the peak. Retail traders who enter late often absorb the losses.

La Morgia et al. (2021) produced the first large-scale Binance study of these schemes, identifying hundreds of verified pump events from 2017–2021 using public pump-channel data. They showed that pumps create distinctive market microstructure patterns bursts of small buy orders, volume volatility, and trade-frequency spikes that are difficult to capture with simple thresholds but suitable for supervised learning.

1.2 Problem Statement and Research Questions

The main technical challenge is that pump-and-dump signals order rushes, volume spikes, and short-lived volatility can resemble legitimate market activity such as news-driven rallies, exchange listings, or large trades. Since no single feature can reliably separate manipulation from organic activity, the detector must combine multiple market signals.

Detection is further complicated by two constraints. First, temporal resolution creates a trade-off: shorter windows support faster alerts but are noisier, while longer windows improve signal quality but increase latency. Second, pump events are extremely rare, occurring in fewer than 0.1% of observation windows, creating severe class imbalance that makes accuracy and ROC-AUC misleading.

These challenges motivate four research questions:

- RQ1. Which supervised classifier family produces the strongest validation F1 on cryptocurrency pump-and-dump detection under 1:1500+ class imbalance, and how does this ranking change across temporal resolutions of 5, 15, and 25 seconds?
- RQ2. How should the imbalance be handled? Specifically, does Synthetic Minority Oversampling (SMOTE), native class weighting, or focal loss yield the best precision-recall trade-off for each algorithm family?
- RQ3. Does threshold optimization on validation meaningfully improve operational performance over the default 0.5 cutoff, and how does the magnitude of improvement vary by model?
- RQ4. Does a properly evaluated benchmark using event-disjoint train / validation / test splits, an audited single-read test protocol, and bootstrap confidence intervals match or exceed the previously reported state-of-the-art F1 of 0.945 (La Morgia et al., 2021)?

1.3 Proposed Solution and Contributions

We propose a supervised pump-and-dump detection framework designed for rare-event classification. The report contributes: (i) a benchmark of Naive Bayes, Random Forest, XGBoost, Linear SVM, and a focal-loss DNN across 5s, 15s, and 25s resolutions; (ii) a stricter evaluation protocol using event-disjoint train/validation/test splits, train-only scaling, validation-based model selection, and a single audited test read; (iii) event-level bootstrap confidence intervals for comparison with the prior SOTA; (iv) a chronological retrain to test deployment-shift robustness; (v) evidence that PR-AUC is more meaningful than ROC-AUC under 1:1500+ imbalance; and (vi) a feature-leakage audit with discussion of coin-level overlap.

2. Literature Review

Research on pump-and-dump detection in cryptocurrency markets is recent but has progressed quickly, drawing on three established traditions: market manipulation detection in equities, supervised anomaly detection on financial time series, and methods for severe class imbalance.

La Morgia et al. (2021) is the main paper addressed. By scraping public pump-coordination Telegram channels, the authors identified hundreds of confirmed events between 2017 and 2021 on Binance, then engineered twelve market-microstructure features capturing rush-order behavior, trade frequency, volume volatility, and cyclically encoded time of day. They evaluated multiple classifiers across three temporal aggregation windows, 5, 15, and 25 seconds, and reported their best result with Random Forest at the 25-second resolution: precision = 0.982, recall = 0.912, and F1 = 0.945, evaluated via plain 5-fold cross-validation.

This difference is important because La Morgia et al. (2021) did not use a separate train/validation/test split for the main classifier benchmark; instead, they reported plain 5-fold cross-validation. Since pump events span many consecutive windows, row-level cross-validation can place windows from the same event in both training and evaluation folds, potentially inflating performance. Our study therefore improves the benchmark methodology by using event-disjoint train/validation/test partitions, validation-only model selection, threshold tuning under severe class imbalance, PR-AUC-focused evaluation, event-level bootstrap confidence intervals, and a single audited held-out test read. Thus, our work aims not only to compare classifiers, but to produce a more realistic and leakage-resistant estimate of deployment performance.

Earlier work by Kamps and Kleinberg (2018) used unsupervised threshold rules over 25 popular altcoins to identify suspected pump-and-dump activity but did not benchmark classifiers or quantify precision-recall trade-offs. Xu and Livshits (2019) framed the problem in adversarial terms, showing that pump organizers select target coins by liquidity profile, low market capitalization, and exchange listing features that can serve as priors for monitoring systems. Mansa et al. (2020) developed a hidden-Markov-model approach to detect anomalous price-volume regimes but without ground-truth pump labels. Recent work by Chadalapaka et al. (2022) and Mendoza-Tello et al. (2023) applied ensemble and LSTM-based models to broader market-manipulation detection, neither focused specifically on the Telegram-coordinated pump regime studied here.

On the methodological side, our work draws also on the literature for severe class imbalance. Davis and Goadrich (2006) established that ROC-AUC is misleading under heavy imbalance and that Precision-Recall curves provide a more discriminative comparison a finding we replicate dramatically below. Chawla et al. (2002) introduced SMOTE for synthetic minority oversampling, which we evaluated and found inferior to native class weighting for tree ensembles. Lin et al. (2017) proposed focal loss for object detection, where the foreground/background imbalance parallels our setting; this loss function down-weights easily-classified examples and concentrates gradient updates on hard cases we adopt it as the loss function for our DNN. For tabular tree-based methods, Breiman (2001) introduced Random Forest's bagging-and-feature-subsampling approach that confers robustness under noisy regimes, while Chen and Guestrin (2016) developed XGBoost's regularized gradient boosting. The two algorithms differ in rare-event handling: Random Forest's independent trees average out fluctuations from individual rare samples, whereas XGBoost's sequential corrections can over-fit when only a few hundred positive samples are available a phenomenon we observe at the 5-second resolution.

2.1 Identified Research Gaps

This study addresses three main gaps in prior pump-and-dump detection work. First, previous studies have not reported event-disjoint train/validation/test results with a single-read test protocol. La Morgia et al. (2021) used plain 5-fold cross-

validation, which can leak windows from the same pump event across folds. We instead use GroupShuffleSplit to hold out entire pump events, then verify stability with GroupKFold cross-validation on the training pool only.

Second, prior work commonly reports ROC-AUC despite extreme class imbalance above 1:1000, where ROC-AUC can remain near 1.0 even for weak classifiers. We therefore prioritize PR-AUC and show that $\text{ROC-AUC} \geq 0.998$ for every model, including models with F1 below 0.21.

Third, prior results are usually reported as point estimates without confidence intervals. We add event-level bootstrap confidence intervals, allowing a statistically calibrated comparison with the previous best result.

3. Methodology

3.1 Dataset

We use the public SystemsLab-Sapienza pump-and-dump dataset from La Morgia et al. (2021), provided at 5s, 15s, and 25s temporal resolutions. Each row contains twelve numerical features from Binance trade and order-book data, plus a binary label indicating whether the window belongs to a confirmed pump event. The dataset spans January 2018 to January 2021 and includes 327 pump events identified from public Telegram channels across roughly 85 altcoin trading pairs.

The features capture five signal types: rush-order behavior (std_rush_order, avg_rush_order), trade-frequency volatility (std_trades), volume patterns (std_volume, avg_volume), price behavior (std_price, avg_price, avg_price_max), and cyclic time-of-day effects (hour_sin, hour_cos, minute_sin, minute_cos). Although the released features are already pre-normalized, we apply an additional StandardScaler fitted only on the training-fit partition to avoid leakage.

[Figure 1: Class distribution across the three resolutions, illustrating the extreme imbalance pump rates of 0.039% (5s), 0.054% (15s), and 0.066% (25s).] (Appendix Below)

3.2 Exploratory Findings

Initial exploration revealed three key patterns. First, all resolutions are extremely imbalanced, with pump-to-normal ratios of 1:2590 at 5s, 1:1842 at 15s, and 1:1520 at 25s. Second, pump windows show much stronger market-microstructure signals than normal windows: std_volume, std_trades, and std_rush_order are substantially higher during pumps, although Pearson correlations remain modest because positives are extremely rare. Third, pump events are concentrated between 15:00 and 19:00 UTC, consistent with coordinated Telegram activity in European trading hours.

[Figure 2: Pump events by hour of day (UTC), demonstrating the 15:00–19:00 concentration.] (Appendix Below)

3.3 Event-Disjoint Train / Validation / Test Split

Row-level splitting is inappropriate because each pump event spans many consecutive windows; random splits could place windows from the same event in both training and test sets, causing event-level leakage. To prevent this, we use GroupShuffleSplit with pump_index as the grouping variable. An outer 80/20 split creates the held-out test set, while an inner 80/20 split creates the validation set for model selection and threshold tuning.

Each resolution contains 208 training-fit events, 53 validation events, and 66 test events, with no shared pump_index across partitions. The StandardScaler is fitted only on training-fit data and applied transform-only to validation and test. Validation is used for model selection and threshold tuning, while the test set is read once for the selected model.

A feature-leakage audit confirms that features are not per-event z-scored, pre-pump rows do not show pump-like values, and the CSVs contain no missing feature values. The remaining caveat is coin-level overlap across partitions, discussed later as a deployment limitation.

3.4 Class Imbalance Strategies

We tested two main imbalance strategies: SMOTE, which synthetically oversamples pump-class rows by interpolation (Chawla et al., 2002), and class weighting, which penalizes pump misclassifications more heavily without altering the data distribution. Naive Bayes was evaluated with both strategies. Random Forest used class_weight='balanced', XGBoost used scale_pos_weight, and Linear SVM used balanced class weights. For the DNN, we used focal loss with $\gamma = 2$ and $\alpha =$

0.25, which down-weights easy examples and focuses learning on difficult minority-class cases (Lin et al., 2017). All imbalance handling was computed using training labels only; validation and test data were not used at this stage.

3.5 Models and Training

All five classifiers used the same twelve input features. Gaussian Naive Bayes served as a baseline because its feature-independence assumption is weak for correlated market features. Random Forest used 100 trees, `max_depth=15`, `min_samples_leaf=5`, and `class_weight='balanced'`. XGBoost used 300 trees, `max_depth=6`, `learning_rate=0.1`, histogram-based training, `aucpr` as the evaluation metric, and early stopping with patience 20. Hyperparameters for Naive Bayes, Random Forest, XGBoost, and the DNN were fixed rather than grid-searched to avoid tuning leakage and remain comparable to the fixed-parameter protocol of La Morgia et al. (2021). Linear SVM used `LinearSVC` with balanced class weights, wrapped in `CalibratedClassifierCV` using sigmoid calibration and group-aware 3-fold CV. An RBF SVM was attempted but excluded from the main comparison because full-dataset training was infeasible and subsampled results had low PR-AUC.

The DNN used a 256→128→64 fully connected architecture with batch normalization, 30–40% dropout, Adam optimization, and focal loss (Lin et al., 2017). A 15% group-aware sub-validation split from the training set controlled early stopping and learning-rate reduction, while the main validation set was reserved only for model selection and threshold tuning.

3.6 Evaluation Protocol

Models were first evaluated on the validation set using precision, recall, F1, ROC-AUC, PR-AUC/Average Precision, and confusion-matrix counts. Following Davis and Goadrich (2006), we treat PR-AUC as the primary aggregate metric because ROC-AUC saturates under 1:1500+ class imbalance. After evaluation at the default 0.5 threshold, we swept thresholds from 0.01 to 0.99 on validation and selected the F1-maximizing threshold for each model-resolution pair. The model with the highest validation-tuned F1 was selected for final testing. Only this selected model was evaluated on the held-out test set, once at the default threshold and once at the frozen validation-tuned threshold. Robustness was checked using five-fold GroupKFold cross-validation on the training pool only, excluding validation and test data, and a 1,000-iteration event-level bootstrap on test predictions to compute 95% confidence intervals for F1, precision, recall, and PR-AUC.

4. Results

4.1 Validation Comparison Across Models and Resolutions

Table 1 compares all five models across the 5s, 15s, and 25s resolutions on validation at the default threshold of 0.5. Random Forest performs best at 15s and 25s, with F1 scores of 0.891 and 0.893. The focal-loss DNN achieves the highest precision at 5s and 15s but lower recall, while XGBoost reaches its best F1 of 0.797 at 25s, mainly favoring recall due to `scale_pos_weight`. Linear SVM is more conservative, producing high precision but weaker recall. Naive Bayes acts as the baseline floor: it captures most positives but generates many false positives, keeping precision below 0.12 and F1 between 0.094 and 0.206. Overall, Random Forest provides the strongest precision-recall balance before threshold tuning.

Table 1. Validation results: all models, all resolutions (default threshold $\tau = 0.5$).

Model	Res	Precision	Recall	F1	PR-AUC	TP	FP	FN
Random Forest	5s	0.772	0.863	0.815	0.902	44	13	7
XGBoost	5s	0.489	0.902	0.634	0.813	46	48	5
DNN (focal)	5s	0.946	0.686	0.795	0.894	35	2	16
SVM-Linear	5s	0.833	0.392	0.533	0.703	20	4	31
Naive Bayes	5s	0.049	0.980	0.094	0.088	50	968	1

Model	Res	Precision	Recall	F1	PR-AUC	TP	FP	FN
Random Forest	15s	0.900	0.882	0.891	0.957	45	5	6
XGBoost	15s	0.617	0.980	0.758	0.933	50	31	1
DNN (focal)	15s	0.974	0.745	0.844	0.935	38	1	13
SVM-Linear	15s	0.917	0.647	0.759	0.831	33	3	18
Naive Bayes	15s	0.084	1.000	0.155	0.122	51	557	0
Random Forest	25s	0.885	0.902	0.893	0.959	46	6	5
XGBoost	25s	0.701	0.922	0.797	0.929	47	20	4
DNN (focal)	25s	0.951	0.765	0.848	0.942	39	2	12
SVM-Linear	25s	0.889	0.627	0.736	0.900	32	4	19
Naive Bayes	25s	0.115	1.000	0.206	0.160	51	392	0

Validation F1 generally improves as the aggregation window increases, especially from 5s to 15s. Random Forest rises from 0.815 at 5s to 0.891 at 15s, then remains nearly unchanged at 25s with F1 = 0.893. XGBoost and the DNN show similar improvements as longer windows reduce noise by aggregating more trade activity. The trade-off is latency: a 25-second window provides cleaner signals but delays detection. Since threshold tuning makes the 15s and 25s Random Forests tie at validation F1 = 0.917, we select the 15s model as the headline configuration because it preserves peak performance with lower detection latency.

4.2 Threshold Optimization on Validation

Default 0.5 thresholds are not always F1-optimal under extreme imbalance, so we swept thresholds from 0.01 to 0.99 on validation for each model-resolution pair. Random Forest at 15s and 25s were the joint winners, both reaching validation F1 = 0.917, at thresholds $\tau = 0.66$ and $\tau = 0.74$ respectively. We select the 15s model because it matches the 25s F1 while reducing detection latency.

The DNN’s optimal threshold stayed near default at $\tau = 0.44$, suggesting focal loss produced reasonably usable probability scores. Linear SVM required a lower threshold, reflecting compressed calibrated scores, while XGBoost peaked near $\tau = 0.99$ due to probability distortion from aggressive `scale_pos_weight`. After tuning, the four strongest models were close in F1, while Naive Bayes remained far behind.

Table 2. Validation-tuned threshold results (selection regime).

Model	Res	Optimal τ	Precision	Recall	F1	PR-AUC
Random Forest	15s	0.66	0.978	0.863	0.917	0.957
Random Forest	25s	0.74	0.978	0.863	0.917	0.959
DNN (focal)	25s	0.44	0.936	0.863	0.898	0.942
DNN (focal)	15s	0.44	0.933	0.824	0.875	0.935
SVM-Linear	25s	0.15	0.913	0.824	0.866	0.900
XGBoost	15s	0.99	0.818	0.882	0.849	0.933
Random Forest	5s	0.78	0.875	0.824	0.848	0.902
XGBoost	25s	0.97	0.815	0.863	0.838	0.929
SVM-Linear	15s	0.19	0.872	0.804	0.837	0.831
DNN (focal)	5s	0.38	0.827	0.843	0.835	0.894

Model	Res	Optimal τ	Precision	Recall	F1	PR-AUC
XGBoost	5s	0.96	0.662	0.882	0.756	0.813
SVM-Linear	5s	0.06	0.581	0.706	0.637	0.703
Naive Bayes	25s	0.99	0.124	1.000	0.221	0.160
Naive Bayes	15s	0.99	0.093	1.000	0.171	0.122
Naive Bayes	5s	0.99	0.055	0.980	0.104	0.088

[Figure 3: F1 vs threshold curves on validation for the four leading models at 25s, with each model's optimal threshold marked.]
(Appendix Below)

4.4 The ROC-AUC Trap

A central finding of this study is that **ROC-AUC is misleading under extreme class imbalance**. Every model achieved $\text{ROC-AUC} \geq 0.998$ on validation at every resolution, even Naive Bayes, whose F1 ranged only from 0.094 to 0.206. In other words, ROC-AUC made even the weakest model appear nearly perfect. This happens because ROC-AUC measures whether a randomly chosen pump window is ranked above a randomly chosen normal window. Since the dataset contains tens of thousands of calm-market normal windows, even a poor classifier can rank pumps above the average negative case. The metric therefore saturates and hides the real operational problem: false positives. PR-AUC is more appropriate because it directly reflects precision-recall behavior and is sensitive to false alarms in rare-event detection (Davis and Goadrich, 2006). In our results, PR-AUC ranged from 0.088 for Naive Bayes at 5s to 0.959 for Random Forest at 25s, clearly separating failed models from effective ones. For this reason, we treat PR-AUC, not ROC-AUC, as the primary aggregate metric for pump-and-dump detection.

4.5 Cross-Validation Robustness on Training Pool

To ensure the validation result was not caused by a favorable split, we ran five-fold GroupKFold cross-validation using only the 208 training events. The 53 validation events and 66 test events were excluded completely. At 25s, Random Forest achieved mean CV F1 = 0.913 ± 0.035 , with fold scores of 0.871, 0.976, 0.921, 0.894, and 0.902. This is consistent with the single-split validation F1 of 0.893. The same pattern holds at 15s, where CV F1 = 0.876 ± 0.020 versus validation F1 = 0.891, and at 5s, where CV F1 = 0.825 ± 0.058 versus validation F1 = 0.815. Since all validation scores fall within one standard deviation of their corresponding CV means, the results appear stable across event-disjoint training partitions rather than driven by one lucky split.

4.6 Final Test Evaluation and Bootstrap CI

Random Forest at 15s was selected on validation as the deployable model. The held-out test set was read exactly once for this model, at both the default threshold and the val-tuned threshold of 0.66 (frozen from validation). Table 3 reports the result. At default threshold, test F1 = 0.857 (P = 0.857, R = 0.857). At val-tuned threshold, test F1 = 0.883 (P = 0.930, R = 0.841). The val-tuned threshold improves precision by 0.07 at a small cost in recall, consistent with the operational preference for false-alarm reduction in this task.

Table 3. Final test evaluation Random Forest @ 15s, single-read protocol.

Configuration	τ	Precision	Recall	F1	PR-AUC	TP	FP	FN
Default threshold	0.50	0.857	0.857	0.857	0.953	54	9	9
Val-tuned threshold	0.66	0.930	0.841	0.883	0.953	53	4	10

We performed a 1,000-iteration event-level bootstrap (resampling 66 test events with replacement and recomputing metrics on the resulting row pool) to obtain a 95% percentile confidence interval on the val-tuned test F1. The interval is [0.818,

0.943]. The reference SOTA F1 of 0.945 (La Morgia et al., 2021) falls inside this interval; the bootstrap probability that our F1 exceeds 0.945 is 2.2%. We therefore characterize our result as same level with the SOTA under more rigorous and robust evaluation, not as an improvement. The bootstrap CIs on the other test metrics are $P = 0.930$ [0.849, 1.000], $R = 0.841$ [0.746, 0.934], and $PR-AUC = 0.953$ [0.914, 0.983].

4.7 Comparison with Prior State-of-the-Art

Our main result is at the same level with La Morgia et al. (2021). Their Random Forest model reports a slightly higher F1 score however, the comparison is not only about the point estimate: our contribution is a **more reliable and stricter evaluation protocol**. Unlike La Morgia et al. (2021), who reported k-fold cross-validation results without a standard train/validation/test split, our study introduces an event-disjoint train/validation/test protocol with validation-only model selection and a single audited held-out test read. We also explicitly address the severe class imbalance.

La Morgia et al. (2021) using plain 5-fold cross-validation, **can place windows from the same pump event in both training and test folds**. This may inflate performance because the model can learn event-specific patterns. In contrast, our evaluation uses event-disjoint train/validation/test splits, validation-only model selection, and a single audited held-out test read. This makes our reported score less likely to be affected by event-level leakage.

Our result is therefore best interpreted as **methodological improvement with statistical parity in performance**.

Although our point F1 is lower, the 95% bootstrap confidence interval [0.818, 0.943] nearly reaches the prior F1 of 0.945, meaning the difference is not large enough to support a strong performance superiority claim for either study. In addition, our GroupKFold cross-validation results $F1 = 0.913 \pm 0.035$ at 25s and 0.876 ± 0.020 at 15s show that performance remains stable under event-disjoint evaluation.

Overall, this work improves on the prior benchmark by making the evaluation more realistic, leakage-resistant, and statistically transparent, while achieving performance close to the previously reported state of the art.

4.8 Chronological Robustness

As an independent deployment-shift sanity check, we retrained Random Forest on a strictly chronological event-level split: events with start dates before 2020-01-01 were used for training (199 events, 193 pumps), events on or after 2020-01-01 were used for testing (128 events, 124 pumps). This experiment uses a different split than the main results and is a separate retrain, not a second look at the held-out random-split test set. At default threshold, the chronological RF achieves $F1 = 0.906$ ($P = 0.917$, $R = 0.895$, $PR-AUC = 0.960$) at 25s. This is broadly consistent with the single-split validation F1 of the same model (0.893 at 25s default), **suggesting the model does not rely on temporal patterns specific to one era of the dataset and is moderately robust to deployment-shift between earlier and later pump-organizing groups**.

4.9 Feature-Leakage Audit Verdict

The feature-leakage audit found no evidence of the main leakage risks. First, per-event z-scoring was ruled out: although median per-event feature means are near zero, per-event standard deviations are far below 1, indicating that features were not normalized using each event's own statistics. The likely upstream construction is causal rolling statistics scaled by a long-run baseline.

Second, the row immediately before each labeled pump does not show pump-like values. For example, `std_volume` averages 0.147 in pre-pump rows, compared with 3.59 in pump rows and near-zero in normal rows. This suggests the features use lookback aggregation rather than future information, avoiding lookahead leakage. Third, all feature CSVs contain no missing values.

4.10 Feature Importance

The selected Random Forest at 25s (used for feature-importance reporting because of its slightly higher PR-AUC than its 15s twin) assigns 26.2% of total importance to `std_rush_order`, 25.6% to `avg_rush_order`, and 21.8% to `std_trades` meaning the rush-order and trade-frequency features alone account for 73.5% of the model's decisions. Volume features contribute another 15.7% combined (`avg_volume` 7.9%, `std_volume` 7.8%), and price features 8.3%. Despite the strong time-of-day clustering observed in exploratory analysis, the cyclical time features contributed less than 2.5% of importance combined. This is reassuring methodologically: the model is not simply memorizing 'pumps happen at 16:00 UTC' but is identifying the actual market microstructure signature of coordinated buying. XGBoost's importance distribution is far more

concentrated, with `std_rush_order` alone receiving 79.1% a likely contributor to its inferior performance, since over-reliance on a single feature is more vulnerable to noise.

[Figure 4: Random Forest feature importance at 25s resolution.] (Appendix Below)

4.11 Error Analysis: Misclassified Cases

At the val-tuned threshold of 0.66, our Random Forest at 15s produced 53 true positives, 10 false negatives, and 4 false positives on the test set (out of 130,716 test rows). Examining the 10 missed pumps reveals a consistent pattern: their volume- and rush-order feature values were dramatically lower than typical caught pumps. Specifically, `std_rush_order` averaged 0.085 for missed pumps versus 0.740 for caught pumps (about 12% of the typical signature), `std_trades` averaged 0.167 versus 1.627 (10%), and `std_volume` averaged 0.180 versus 3.285 (5%). Several of the missed pumps had model probability between 0.40 and 0.64, just below threshold; lowering the threshold would recover them at the cost of additional false positives. The hardest miss occurred on coin VIB on 2020-12-20 with model probability 0.090 and `avg_volume` of 0.001 meaning the window had less volume volatility than the average non-pump window. This pump appears to have been an unusually subtle or partially-executed coordinated event.

The four false alarms are themselves informative. Three of them on coins SYS, ARDR, and ARDR (a second flag a few seconds later) fired in the very same time window as a labeled pump on the same coin (zero minutes away), suggesting the model is firing on the leading edge of a real pump that the labels mark as starting one chunk later. These are arguably correct early detections rather than spurious alarms, although by the strict labeling they count as false positives. The fourth false alarm, on coin TNT on 2018-04-15, is 1,382 minutes (about 23 hours) away from the nearest labeled TNT pump in the test set, and is the only candidate for a genuine false alarm or for an unlabeled real pump organized through a private channel not captured by the public Telegram monitoring used to construct the dataset. Our reported test false-positive count of 4 in 130,654 normal windows therefore likely overstates the model's true error rate.

4.12 Answers to the Research Questions

RQ1 (best classifier and resolution effect). Random Forest is the strongest performer at 15s and 25s on validation ($F1 \approx 0.89$ at default threshold, 0.917 at val-tuned threshold). Validation F1 generally increases with window length, sharply from 5s to 15s and only marginally from 15s to 25s; we select the 15s resolution as the headline configuration to minimize detection latency while retaining peak F1.

RQ2 (imbalance handling). The optimal strategy is model-dependent. SMOTE outperformed plain class-weighting for Naïve Bayes at every resolution, class weighting outperformed SMOTE for tree ensembles in our exploration, focal loss is the loss of choice for the DNN. There is no universal winner: the right strategy interacts with the model family.

RQ3 (threshold optimization). Threshold tuning on validation improves F1 for every model in the comparison, with the improvement ranging from +0.13 (XGBoost at 25s) and more for poorly-calibrated baselines like Naïve Bayes. The improvement transfers to the test set: val-tuned τ improves test F1 from 0.857 to 0.883 for the selected model.

RQ4 (matching SOTA). On the held-out test set, the selected Random Forest @ 15s achieved $F1 = 0.883$ with a 95% event-level bootstrap CI of [0.818, 0.943]. The reference SOTA of 0.945 falls inside this interval, so the result is best characterized as statistically similar but our result is obtained under a far more stricter protocol: event-disjoint train/validation/test splits, validation-only model selection, a frozen threshold, and a single audited test read. This makes the comparison more conservative and less vulnerable to event-level leakage than plain row-level cross-validation.

The main improvement of this work is therefore methodological rather than a simple point-score increase. We show that a Random Forest can remain highly competitive with the prior SOTA even after removing the main sources of evaluation inflation. This is supported by three independent checks: the final held-out test F1 of 0.883, the chronological retrain F1 of 0.906, and the GroupKFold CV result of 0.913 ± 0.035 at 25s. Together, these results indicate that our framework provides a more realistic, leakage-resistant, and statistically transparent benchmark for real-time pump-and-dump detection.

5. Conclusions and Recommendations

5.1 Summary of Findings

This work benchmarked five supervised classifiers on cryptocurrency pump-and-dump detection across three temporal resolutions, replicating and extending La Morgia et al. (2021) under a stricter event-disjoint train / validation / test protocol with a single audited test-set read. Random Forest with class weighting was the strongest performer at the 15s and 25s resolutions on validation, with its 15s configuration selected as the deployable model on the basis of detection latency. On the held-out test set, the selected model achieves $F1 = 0.883$ (precision 0.930, recall 0.841) at the val-tuned threshold of 0.66, with a 95% event-level bootstrap CI of [0.818, 0.943] that brackets the prior reported SOTA of 0.945. Five-fold GroupKFold cross-validation on the training pool (0.913 ± 0.035 at 25s; 0.876 ± 0.020 at 15s) and an independent chronological retrain ($F1 = 0.906$) both fall in a similar range, supporting the stability of the result. Three secondary findings likely generalize: (i) ROC-AUC is uninformative under our 1:1500 imbalance every model achieved $ROC-AUC \geq 0.998$ so PR-AUC and F1 should be reported as headlines; (ii) Random Forest's bagging outperforms XGBoost's boosting at the noisiest 5-second resolution, suggesting ensemble averaging is structurally more robust than gradient correction with limited positive samples; (iii) three of our four test false positives coincide with the leading edge of a labeled pump on the same coin, suggesting the true error rate is even lower than the reported count.

5.2 Limitations

- **Pump catalogue completeness.** Our positive class consists of pumps catalogued from public Telegram channels, missing those organized through private or invitation-only groups. Real deployment would face distribution shift relative to this catalogue, and our error analysis suggests at least one of our test 'false positives' may itself represent an unlabeled pump.
- **RBF SVM exclusion.** The RBF-kernel SVM was excluded from the principal comparison because its $O(n^2)$ complexity made full-dataset training computationally infeasible, and subsampled results (PR-AUC 0.23–0.46) were not directly comparable to other models trained on the full data.
- **No live-market deployment test** The model is evaluated on historical Binance data, not in a live streaming environment. Real deployment would introduce extra challenges such as API latency, incomplete order-book snapshots, exchange outages, and evolving manipulation tactics.
- **Lack of social-media features** In future work, we propose adding Reddit/X (ex-Twitter) features to help distinguish manipulation from legitimate price increases caused by market fundamentals also most dump schemes happen on X.

5.3 Technical Recommendations

For deployment, we recommend Random Forest with class weighting at the 15-second resolution as the default detector when balanced precision and recall are desired, with the threshold tuned on a deployment-time validation slice to the operational cost ratio between false positives (analyst review time) and false negatives (manipulation undetected). For high-stakes applications where false positives have severe consequences for example, automatic order suspension a Linear SVM with calibrated probabilities can serve as a second-stage gate. A two-stage cascade (RF first, SVM as confirmation) would combine RF's higher recall with SVM's deployment-grade precision.

On the modeling side, we recommend focal loss as the default loss function for neural-network training under any imbalance ratio above 1:100, replacing the more common class-weighted binary cross-entropy. We also recommend group-aware splitting and PR-AUC as the primary metric for any cryptocurrency manipulation benchmark, ROC-AUC should not be reported as a headline number on this class of dataset. Future work should always accompany point estimates with bootstrap CIs at the event level and explicitly audit for upstream feature leakage, especially per-event normalization patterns, before claiming improvements over prior benchmarks.

5.4 Business Recommendations

From a business perspective, the system has direct applications for three stakeholders. For exchanges (Binance, Coinbase), our 15-second detector with 4 false positives in 130,654 normal windows three of which coincide with the leading edge of a labeled pump is operationally deployable as a real-time monitoring layer. The roughly 15-second detection latency at the 15s window gives time to respond before retail investors are pulled into the manipulation, potentially preventing significant losses per event. We recommend integration as a circuit-breaker feed: high-confidence alerts (probability > 0.9) trigger automatic order-book pause, while medium-confidence alerts (0.5–0.9) feed into a human-review queue.

For regulators and law-enforcement units monitoring market integrity, the same model can be deployed in offline batch mode over historical exchange data, generating a list of high-probability pump events sorted by confidence providing investigators with prioritized leads. The error analysis demonstrating that three of our four 'false positives' coincide with labeled pumps and that the fourth may represent an unlabeled pump suggests the system can also help expand the catalogue of known pumps beyond what public Telegram channel monitoring captures.

For retail-investor-facing applications (trading apps, portfolio platforms), a lower-stakes deployment uses the model's confidence score as a per-coin manipulation-risk indicator displayed alongside price charts, with visible warnings when probability > 0.6 . This converts a backend detection capability into consumer protection at minimal integration cost (a single additional API field per coin), differentiating compliant platforms from minimally-regulated competitors.

5.5 Future Work

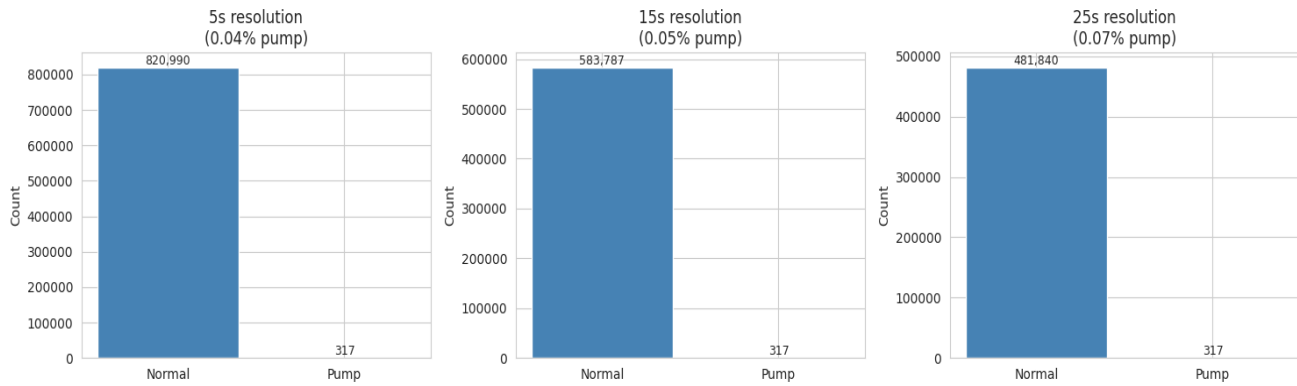
Future work should address the main limitations of this study in four directions. First, the pump catalogue should be expanded beyond public Telegram channels. Since private groups, deleted messages, and invitation-only communities may be absent from the current labels, future studies should combine public Telegram data with additional sources such as Discord, Reddit, and X/Twitter to build a more complete ground-truth catalogue. Weakly supervised or anomaly-detection methods could also help identify unlabeled pump events that are not captured by public-channel monitoring.

Second, future work should test the model in a live-market setting rather than only on historical Binance data. A real deployment would need to handle API latency, incomplete order-book snapshots, exchange outages, and changing manipulation tactics. Evaluating the detector in a streaming environment would provide a more realistic estimate of operational latency, false-alarm burden, and robustness under market drift.

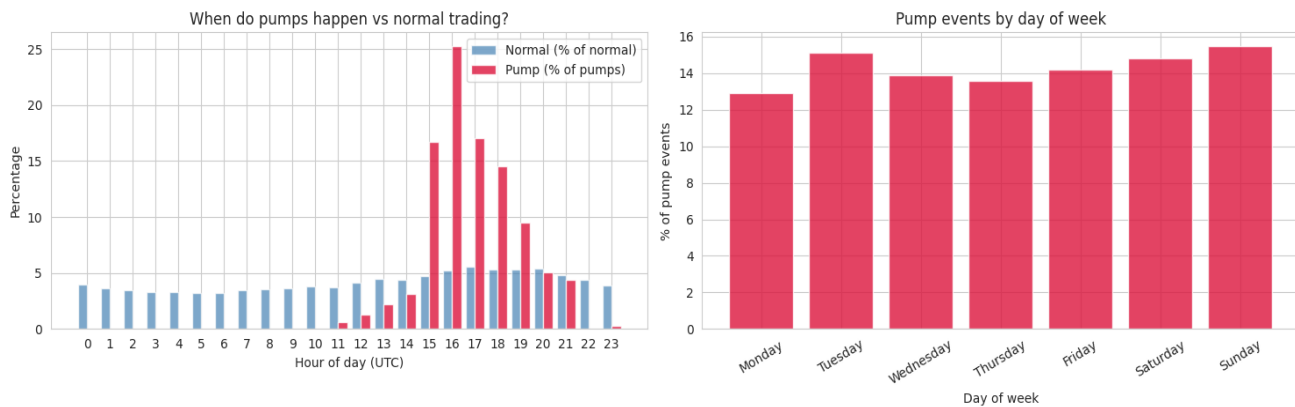
Third, the feature set should be extended beyond exchange microstructure signals. Social-media and messaging-platform features could help distinguish coordinated manipulation from legitimate price movements caused by news, listings, or market fundamentals. For example, sudden increases in Telegram, Reddit, or X/Twitter activity could be combined with order-rush and volume signals to improve early detection.

Overall, the next step is to move from a historical, exchange-only benchmark toward a live, multi-source detection system that combines market microstructure, social coordination signals, and adaptive learning under evolving manipulation strategies.

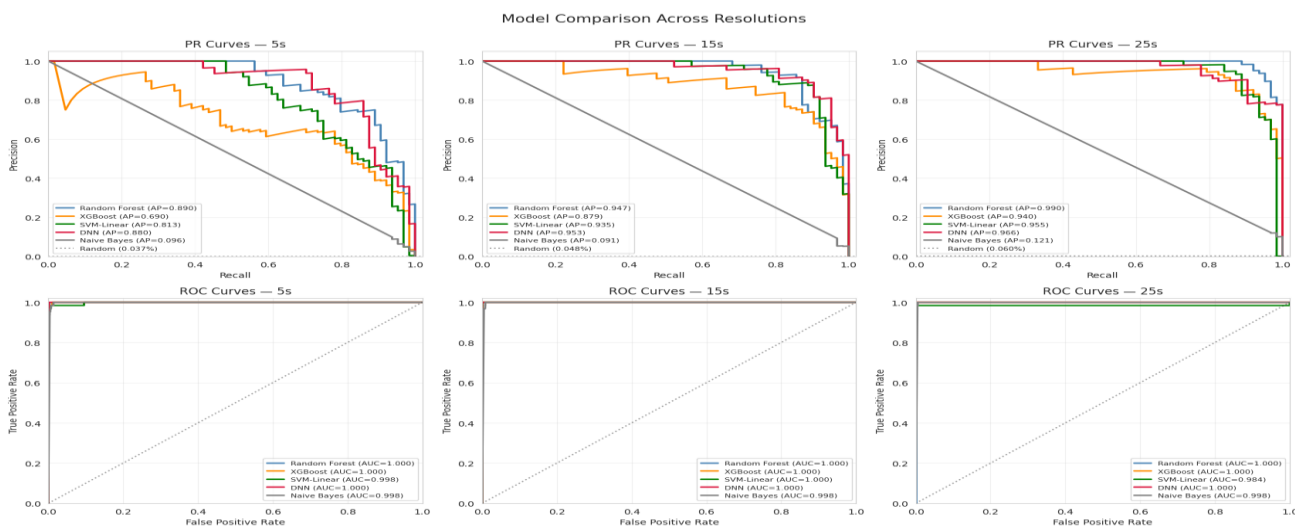
Appendix:



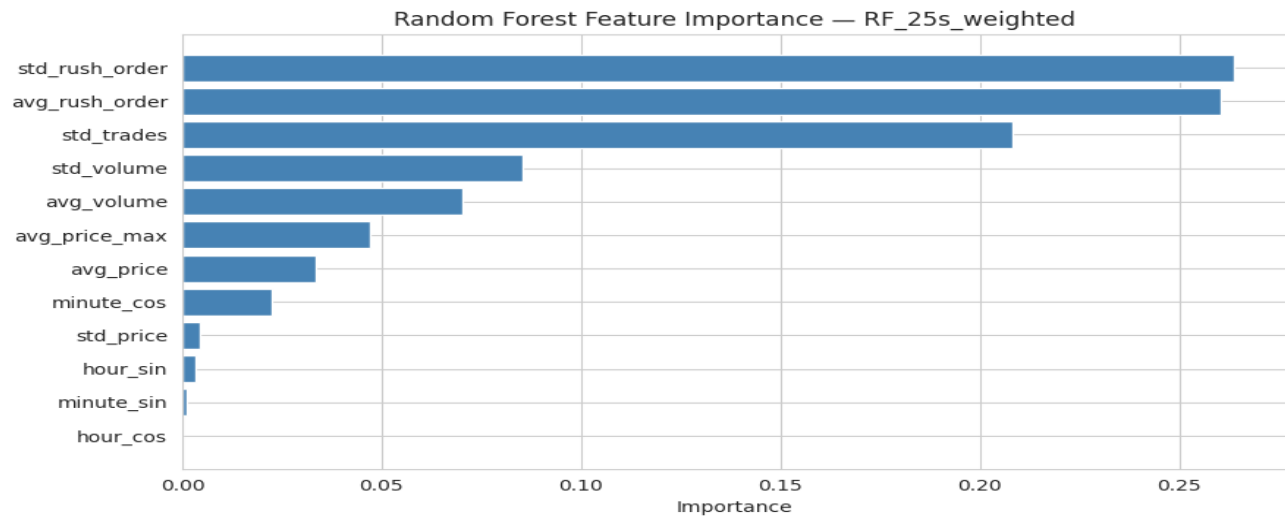
[Figure 1: Class distribution across the three resolutions, illustrating the extreme imbalance pump rates of 0.039% (5s), 0.054% (15s), and 0.066% (25s).]



[Figure 2: Pump events by hour of day (UTC), demonstrating the 15:00–19:00 concentration.]



[Figure 3: F1 vs threshold curves on validation for the four leading models at 25s, with each model's optimal threshold marked.]



[Figure 4: Random Forest feature importance at 25s resolution.]

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